

UNO Extended Edition

REST API Specification — Etapa #1

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1 Overview

1.1 Descrierea problemei

UNO este unul dintre cele mai populare jocuri de cărți din lume. Scopul este să fii primul jucător care își golește mâna, potrivit cărților după color sau type. Proiectul definește un REST API pentru o **versiune extinsă** a jocului UNO, care include house rules configurabile: card stacking, hand swapping (regula 7-0), jump-in, speed mode, team play.

API-ul gestionează întregul game lifecycle: user registration, game creation și configuration, player management, card dealing, turn-based gameplay, rule enforcement, scoring on multiple rounds și global leaderboards.

1.2 Arhitectură generală

Sistemul urmează o arhitectură standard pe 3 layers:

- **Client Layer** — Aplicație browser care utilizează REST API-ul.
- **API Layer** — Stateless RESTful service, built cu **PHP (Slim Framework)**, descris folosind **Swagger 2.0 (OAS 2.0)**. Handle-uieste HTTP requests, JWT authentication, input validation
- **Data Layer** — Bază de date relațională (MySQL/MariaDB) care persistă users, games, cards, turns și scores.

1.3 Actori

Actor	Descriere
Guest	User neautentificat; poate vedea open games și leaderboard-ul
Player	User autentificat; poate crea/intra în jocuri
Game Host	Player-ul care a creat meciul; configurează reguli
Spectator	User autentificat care urmărește jocul
Admin	System administrator; manage-uieste users, bans, moderation

1.4 Tehnologii

Componentă	Tehnologie
Backend	PHP 8+ cu Slim Framework
API Specification	Swagger 2.0 (OAS 2.0, YAML)
Editor	Swagger Editor
Documentation	Swagger UI
Database	MySQL / MariaDB
Authentication	JWT (Bearer token în Authorization header)
Presentation	L ^A T _E X

2 Resources și Endpoints

2.1 Resource Overview

Resource	Route	Tip
User	/users/{userId}	Document
PlayerStats	/users/{userId}/stats	Document
Game	/games/{gameId}	Document
GameSettings	/games/{gameId}/settings	Document
Player	/games/{gameId}/players/{playerId}	Document
Hand	/games/{gameId}/players/{playerId}/hand	Document
DrawPile	/games/{gameId}/draw-pile	Document
DiscardPile	/games/{gameId}/discard-pile	Document
CurrentTurn	/games/{gameId}/turns/current	Document
Round	/games/{gameId}/rounds/{roundNumber}	Document
Scoreboard	/games/{gameId}/scoreboard	Document
Leaderboard	/leaderboard	Document
Games (list)	/games	Collection
Players (list)	/games/{gameId}/players	Collection
Turns (list)	/games/{gameId}/turns	Collection
Rounds (list)	/games/{gameId}/rounds	Collection
Penalties (list)	/games/{gameId}/penalties	Collection

2.2 Users

GET	/users/{userId} <i>Get user profile</i>
Parameter	
userId	UUID-ul user-ului (path param)
Response application/json	
200 — OK Profile returnat	
<pre>{ "id": "uuid", "email": "user@example.com", "displayName": "Player1", "avatarUrl": null, "createdAt": "2026-03-01T10:00:00Z" }</pre>	
404 — Not Found User-ul nu există	

GET	/users/{userId}/stats <i>Get player lifetime statistics</i>
Parameter	
userId	UUID-ul user-ului (path param)
Response application/json	
200 — OK Stats returnate	
<pre>{ "gamesPlayed": 42, "gamesWon": 18, "winRate": 0.43, "totalPoints": 5240, "unoCallsMade": 31, "challengesWon": 7 }</pre>	
404 — Not Found User-ul nu există	

2.3 Leaderboard

GET	/leaderboard <i>Get global player rankings (public, no auth required)</i>
Parameter	
sortBy	Sort field: wins, winRate, totalPoints (query param)
page	Page number, default 1 (query param)
pageSize	Results per page, default 20, max 100 (query param)
Response	application/json
200 — OK	Leaderboard entries
<pre>{ "entries": [{ "rank": 1, "displayName": "UnoMaster", "gamesWon": 120, "winRate": 0.67, "totalPoints": 24500 }], "totalCount": 843, "page": 1, "pageSize": 20 }</pre>	

2.4 Games

GET	/games <i>List all games</i>
Parameter	
status	Filter după phase: lobby, playing, finished (query param)
maxPlayers	Filter după max player count (query param)
Response	application/json
200 — OK	Lista de games
<pre>[{ "id": "uuid", "name": "Friday Night UNO", "phase": "lobby", "maxPlayers": 6, "pointsToWin": 500 }]</pre>	

POST	/games <i>Create a new game</i>
Body	application/json
<pre>{ "name": "Friday Night UNO", "maxPlayers": 6, "pointsToWin": 500, "settings": { "allowStacking": true, "sevenZeroRule": true } }</pre>	
Response	application/json
201 — Created Game creat 400 — Bad Request Invalid configuration	

GET	/games/{gameId} <i>Get full game state</i>
Parameter	
gameId	UUID-ul game-ului (path param)
Response	application/json
200 — OK Game state complet <pre>{ "id": "uuid", "name": "Friday Night UNO", "phase": "playing", "playDirection": "clockwise", "activeColor": "red", "currentRound": 2, "pendingDrawCount": 0 }</pre>	
404 — Not Found Game-ul nu există	

GET	/games/{gameId}/settings <i>Get house rules configuration</i>
Parameter	
gameId	UUID-ul game-ului (path param)
Response application/json	
200 — OK	Game settings
<pre> { "allowStacking": true, "sevenZeroRule": true, "allowJumpIn": false, "drawUntilPlayable": false, "speedMode": false, "teamMode": false, }</pre>	
404 — Not Found	Game-ul nu există

2.5 Players

GET	/games/{gameId}/players <i>Get all players dintr-un game</i>
Parameter	
gameId	UUID-ul game-ului (path param)
Response application/json	
200 — OK	Player list
<pre> [{ "id": "uuid", "displayName": "Player1", "cardCount": 5, "totalScore": 120, "saidUno": false, "isActive": true }]</pre>	

POST	/games/{gameId}/players <i>Join a game</i>
Parameter	
gameId	UUID-ul game-ului (path param)
Response application/json	
201 — Created Joined cu succes	
400 — Bad Request Game full sau deja started	
409 — Conflict Deja în acest game	

2.6 Hand și Piles

GET	/games/{gameId}/players/{playerId}/hand <i>Get player's hand (private, doar own hand)</i>
Parameter	
gameId	UUID-ul game-ului (path param)
playerId	UUID-ul player-ului (path param)
Response application/json	
200 — OK Cards din hand	
<pre> { "cards": [{"id": "uuid", "color": "red", "type": "number", "value": 7}, {"id": "uuid", "color": null, "type": "wild", "value": null}], "count": 2 } </pre>	
403 — Forbidden Nu poți vedea hand-ul altui player	

GET	/games/{gameId}/discard-pile <i>Get discard pile (top card + active color)</i>
Parameter	
gameId	UUID-ul game-ului (path param)
Response application/json	
200 — OK	Discard pile state
<pre> { "topCard": { "id": "uuid", "color": "blue", "type": "skip" }, "activeColor": "blue", "cardCount": 23 } </pre>	

GET	/games/{gameId}/draw-pile <i>Get draw pile card count (cards sunt hidden)</i>
Parameter	
gameId	UUID-ul game-ului (path param)
Response application/json	
200 — OK	Draw pile info
<pre> { "cardCount": 45 } </pre>	

2.7 Turns, Rounds, Scoreboard, Penalties

GET	/games/{gameId}/turns/current <i>Get current turn info</i>
Parameter	
gameId	UUID-ul game-ului (path param)
Response	application/json
200 — OK	Current turn state
<pre>{ "currentPlayerId": "uuid", "playDirection": "clockwise", "turnNumber": 15, "roundNumber": 2, "pendingDrawCount": 0 }</pre>	

GET	/games/{gameId}/turns <i>Get turn history</i>
Parameter	
gameId	UUID-ul game-ului (path param)
roundNumber	Filter după round (query param)
playerId	Filter după player (query param)
action	Filter după action type (query param)
Response	application/json
200 — OK	Array de Turn objects

GET	/games/{gameId}/scoreboard <i>Get cumulative scoreboard</i>
Parameter	
gameId	UUID-ul game-ului (path param)
Response	application/json
200 — OK	Scoreboard
<pre> { "players": [{ "playerId": "uuid", "displayName": "Player1", "totalScore": 320, "roundScores": [120, 200] }] }</pre>	

GET	/games/{gameId}/penalties <i>Get penalties dintr-un game</i>
Parameter	
gameId	UUID-ul game-ului (path param)
playerId	Filter după player (query param)
type	Filter după penalty type (query param)
Response	application/json
200 — OK	Array de Penalty objects

3 Controller Actions

Aceste actions reprezintă game mechanics care nu se mapează pe standard CRUD operations. Sunt implementate ca POST endpoints sub `/games/{gameId}/actions/`.

POST	<code>/games/{gameId}/actions/start</code> <i>Start game (host only)</i>	
Response		application/json
200 — OK Game started, first round begins		
400 — Bad Request Not enough players (min 2)		
403 — Forbidden Not the host		

POST	<code>/games/{gameId}/actions/play-card</code> <i>Play a card din hand</i>	
Body		application/json
<pre>{ "cardId": "uuid", "chosenColor": "green" }</pre>		
Response		application/json
200 — OK Card played cu succes		
<pre>{ "success": true, "action": "play_card", "nextTurnPlayerId": "uuid" }</pre>		
400 — Bad Request Card-ul nu match-uește sau nu e în hand		
403 — Forbidden Not your turn		

POST	/games/{gameId}/actions/draw-card <i>Draw card(s) din draw pile</i>
Response application/json	
200 — OK	Card(s) drawn
<pre>{ "success": true, "action": "draw_card", "nextTurnPlayerId": "uuid" }</pre>	
403 — Forbidden	Not your turn

POST	/games/{gameId}/actions/say-uno <i>Declară UNO (un singur card rămas)</i>
Response application/json	
200 — OK	UNO declared
400 — Bad Request	Mai mult de un card în hand

POST	/games/{gameId}/actions/choose-color <i>Choose color după Wild/Wild+4</i>
Body application/json	
<pre>{ "color": "red" }</pre>	
Response application/json	
200 — OK	Color setat
400 — Bad Request	No color choice pending

POST	/games/{gameId}/actions/end-turn <i>End current turn</i>
Response	application/json
200 — OK	Turn ended
403 — Forbidden	Not your turn
409 — Conflict	Trebuie să draw sau să play întâi

4 Data Models

4.1 Enumerations

Enum	Values
CardColor	red, yellow, green, blue
CardType	number, skip, reverse, draw_two, wild, wild_draw_four
GamePhase	lobby, playing, round_over, finished
PlayDirection	clockwise, counter_clockwise
GameAction	play_card, draw_card, say_uno, challenge_uno, challenge_wild_draw_four, choose_color, end_turn, stack_draw, swap_hands, pass_all_hands, jump_in
PenaltyType	missed_uno, illegal_wild_draw_four, timeout, forfeit

4.2 Card Schema

Field	Type	Required	Descriere
id	UUID	da	Unique card identifier
color	CardColor	nu	Null pentru wild cards
type	CardType	da	Card type
value	integer (0–9)	nu	Face value doar la number cards

4.3 Deck Composition

Standard deck (108 cards):

- 76 number cards: câte două de la 1–9 per color + un 0 per color
- 24 action cards: câte două Skip, Reverse, Draw Two per color
- 8 wild cards: patru Wild + patru Wild Draw Four

4.4 Scoring

Card	Points
Number cards (0–9)	Face value
Skip / Reverse / Draw Two	20 points fiecare
Wild / Wild Draw Four	50 points fiecare

Când un player rămâne fără cărți, câștigă runda și primește puncte egale cu suma cărților ramase în mâinile celorlalți jucători. Jocul se termină când un jucător atinge scorul de câștig predefinit în regulile inițiale (default: 500).

5 House Rules

Următoarele reguli pot fi configurate de lobby host per game prin `GameSettings`:

Rule	Key	Descriere
Stacking	<code>allowStacking</code>	+2 se poate stack-ui pe +2, +4 pe +4. Cumulative draw trece la next player care nu poate stack-ui.
Seven-Zero	<code>sevenZeroRule</code>	Jucarea unui 7 permite swap hands cu orice player. Jucarea unui 0 forțează toți să pass hands în current direction.
Jump-In	<code>allowJumpIn</code>	Joci carte identică out of turn. (acoperi)
Draw Until Playable	<code>drawUntilPlayable</code>	Draw cards până găsești una playable.
Force Play	<code>forcePlayDrawn</code>	Drawn card trebuie jucată dacă e jucabilă.
No Bluff Wild+4	<code>noBluffWildDrawFour</code>	Wild+4 jucabil oricând.
Points Mode	<code>pointsMode</code>	Multi-round play către un point target.
Team Mode	<code>teamMode</code>	Team play (2v2 sau 3v3).
Speed Mode	<code>speedMode</code>	Turn timer enabled; auto-draw la timeout.

6 Use Cases

6.1 Game Lifecycle

1. Host-ul creează un game, configurează house rules și max players.
2. Players intră în lobby.
3. Host-ul pornește meciul; first round începe automat.
4. Fiecare round: se împart 7 cards per player, se pune pe masă top card din draw pile.
5. Playerii joacă pe rând: play a matching card, draw, sau use special actions.
6. Runda se încheie când un player își golește hand-ul.
7. Winner-ul primește points din remaining cards ale oponentilor.
8. New round începe până un player atinge point target-ul.
9. Meciul se termină; punctele finale sunt înregistrate în leaderboard.

6.2 Response Codes Summary

Operation	Success	Common Errors
GET resource	200 OK	404 Not Found
GET collection	200 OK	400 Bad Request
POST create	201 Created	400, 409 Conflict
POST action	200 OK	400, 403 Forbidden, 409 Conflict
Auth required	—	401 Unauthorized